

Option D: same effect as Option B.

23

C

D

Since the net magnetic field at d from wire Y is zero, I_X and I_Y flow in opposite directions.

$$B \propto I \text{ and } B \propto \frac{1}{r}$$

At d away from wire Y,

$$\frac{I_X}{(L+d)} = \frac{I_Y}{(d)}$$

$$\frac{4}{L+d} = \frac{1}{d}$$

$$L = 3d$$

24

D

A